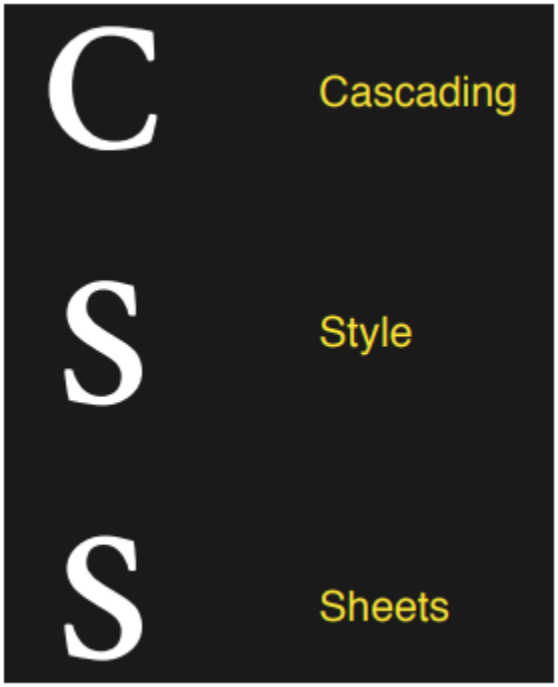
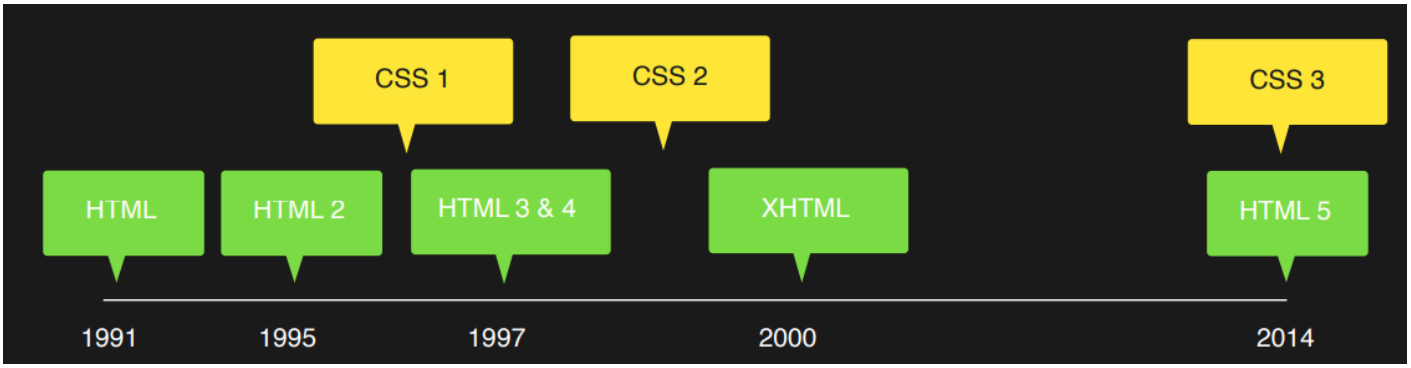
S.P.M College,Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1) **– Hira Kumar**

Web Development





HIRA KUMAR

❖ Properties

- Properties are the CSS attributes that define how an HTML element should be styled.
- Each CSS rule consists of one or more properties followed by a colon (:) and their corresponding values.

Important Properties & value in CSS

1. Color Property

- Color property is used to change the text color
- Example :

  (i) <p style =”color: red;”> Hello world ! </p> {inline} **OR**

  (ii) p{
  Color:red;
  } // internal

```
<head>
<title>hira css</title>
<style>
  p{
    color: red;
  }
</style>
<body>
<p> Hello world !</p>
</body>
```

http://127.0.0.1:5500/index.html

Hello world !

Property	Value
Color:	Red; → Named colors #ff0000; → hexadecimal rgb(255,0,0); → RGB rgba(255,0,0,0.1) → Extension of RGB's, include alpha for opacity control(0-1) range. (where 0.1→ fully transparent, 1.0 → fully opaque) transparent; → transparent currentcolor (current text color)

More Rgb color code / hexadecimal code find, then search **color picker** on browser.

2. Padding Property

- Padding property is used to create space inside the element, between the content and the border. (content और border के बीच अंदर की जगह(space) बनाने के लिए) |
- Syntax : padding: position;
  (position → top, right, bottom, left)

Property	Value	work
Padding:	10px;	All sides
Padding:	10px 20px;	Top/bottom | left/right
Padding:	10px 20px 30px;	Top | left/right | bottom
Padding:	10px 20px 30px 40px;	Top|right|bottom|left

Padding Values

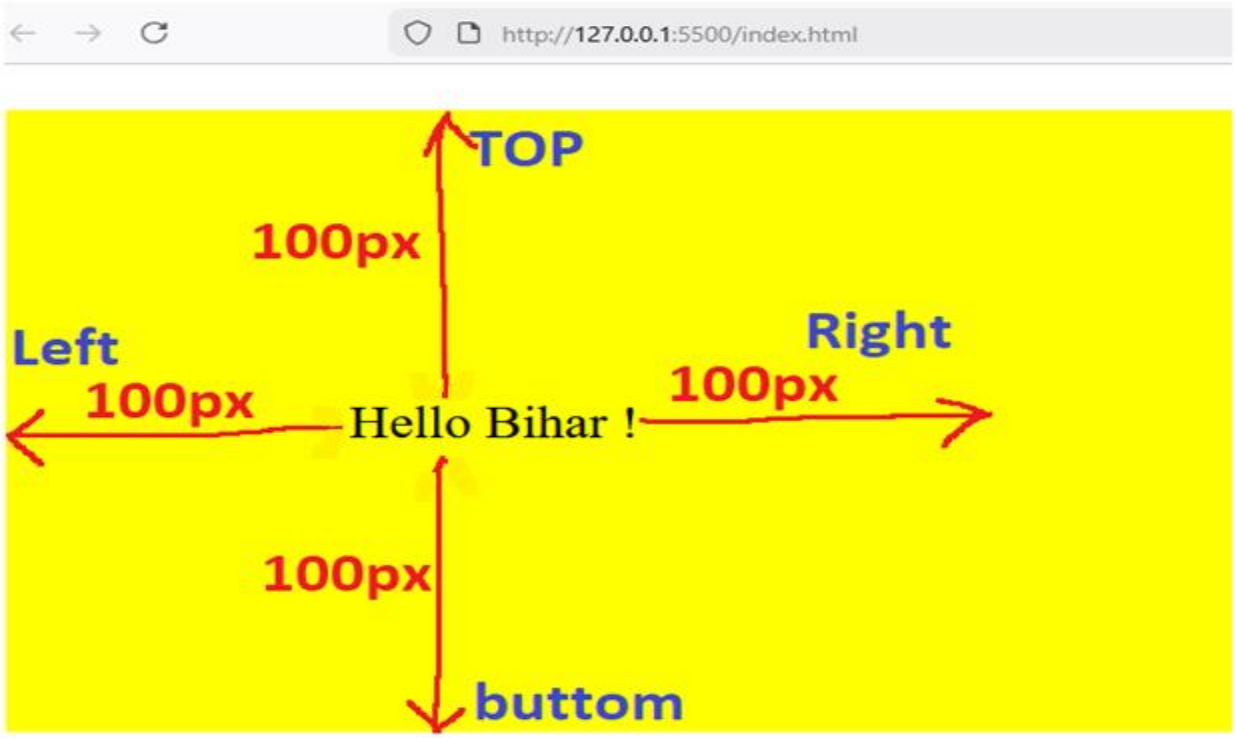
Length Values		
Property	**Value**	**Work**

Padding:	20px;	Pixels
Padding:	1em;	Relative to font-size
Padding:	1rem;	Relative to root font-size
Padding:	5%;	In precentage

Individual side properties and value	
Property	**Value**
Padding-top:	20px;
Padding-right:	10px;
Padding-bottom:	10px;
Padding-left:	25px;

```html
<title>hira css</title>
<style>
  div p{
    background-color : yellow;
   padding:100px;
  }

</style>
<body>
  <div>
    <p> Hello Bihar !</p>
    </div>
    <div>
      <a> Learn HTML & CSS</a>
    </div>
```



Learn HTML & CSS

3. Margin Property

- Margin Property is used to create space outside the element.
- Syntax → margin: position;

  (position = top, right, bottom, left)

Property	Value	work

margin:	10px;	All sides
margin:	10px 20px;	Top/bottom | left/right
margin:	10px 20px 30px;	Top | left/right | bottom
margin:	10px 20px 30px 40px;	Top|right|bottom|left

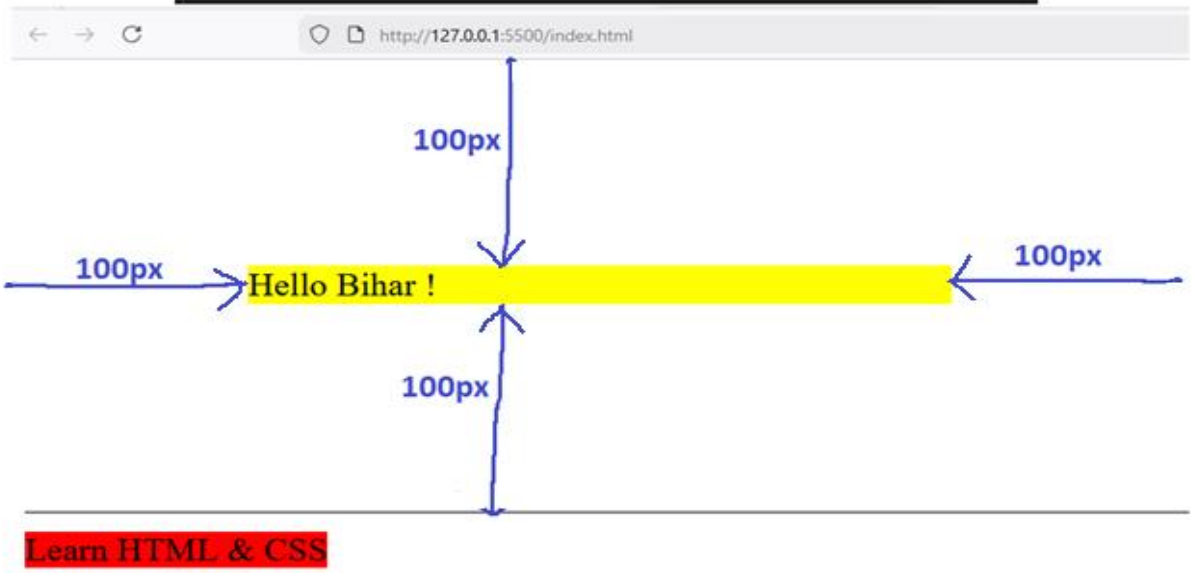
Margin Values

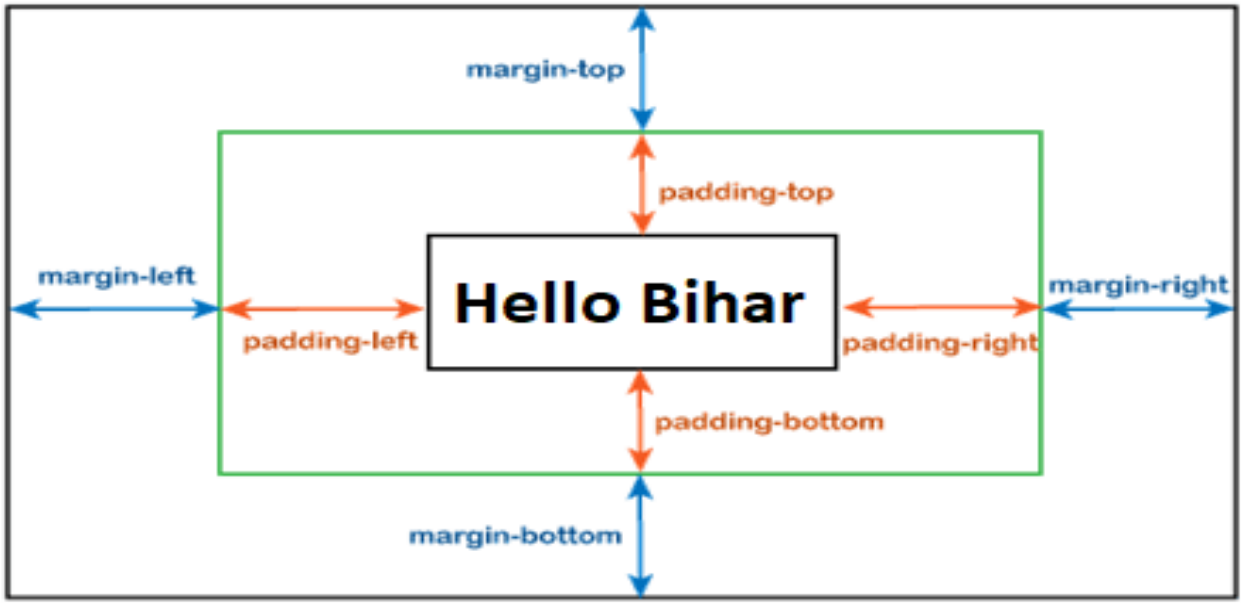
Length Values		
Property	**Value**	**Work**
Margin:	20px;	Pixels
Margin:	1em;	Relative to font-size
Margin:	1rem;	Relative to root font-size
Margin:	5%;	In precentage

Auto Values		
Property	**Value**	**Work**
Margin:	Auto;	Browser calculates the margin
Margin:	0 auto;	Center block elements horizontally

Individual side properties and value	
Property	**Value**
Margin-top:	20px;
Margin-right:	10px;
Margin-bottom:	10px;
Margin-left:	25px;

```html
<title>hira css</title>
<style>
  div p{
    background-color : yellow;
  margin:100px
  }
</style>
<body>
  <div>
    <p> Hello Bihar !</p>
    </div><hr>
    <div>
      <a> Learn HTML & CSS</a>
    </div>
```





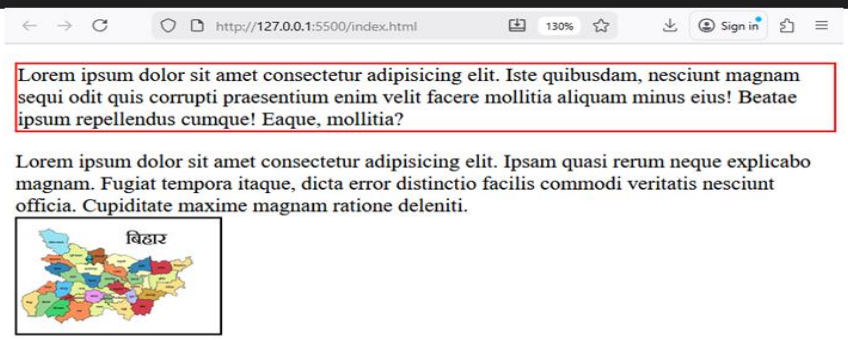
4. Border Property

- Border property is used to create a border(line) around an element.
- Syntax→ border : <border-width> <border-style> <border-color>;

```
<head>
<title>hira css</title>
<style>
    .bihar{
     background-image : url(bihar.jpg);
     background-repeat :no-repeat;
      background-size:contain;
     width:150px;
   height:100px;
   border : 2px solid black;
   }
   p{
     border : 2px solid red;
   }
</style>
<body>

<p> Lorem ipsum dolor sit amet consectetur adipisicing elit. Iste quibusdam, nesciunt magnam sequi odit quis corrupti
praesentium enim velit facere mollitia aliquam minus eius! Beatae ipsum repellendus cumque! Eaque, mollitia?</p>
<a> Lorem ipsum dolor sit amet consectetur adipisicing elit. Ipsam quasi rerum neque explicabo magnam. Fugiat tempora
itaque, dicta error distinctio facilis commodi veritatis nesciunt officia. Cupiditate maxime magnam ratione deleniti.</a>
<div class="bihar">

</div>
```



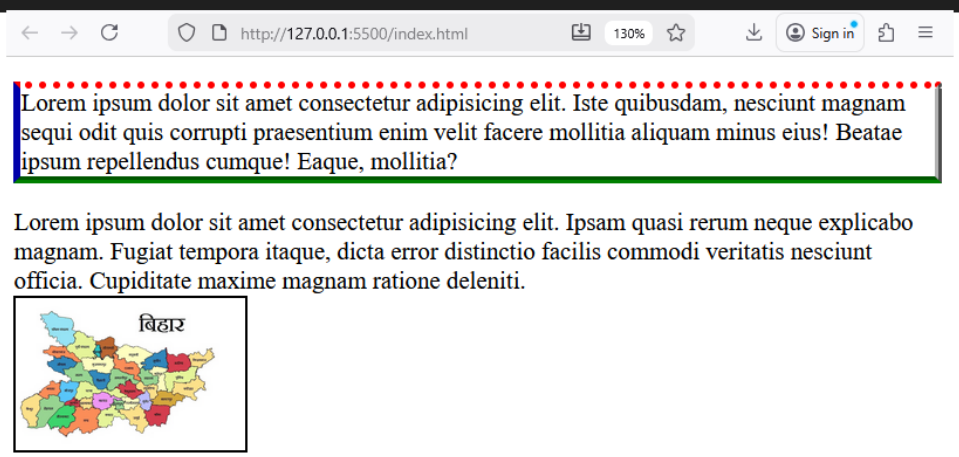
Property	Value
Border-widht:	Thin → 1px medium → 3px, thick→5pc, 10px →custome, 1em,2em,3em,4em → top,right,bottom,left
Border-style:	None; → No border Hidden; → same as none, but for tables Dotted; → dotted border Dashed ; → dashed border Solid; → solid border Double; → double border Groove; → 3d grooved effect Ridge;→ 3d ridge effect Insert → 3d insert effect Outset → 3d outset effect
Border-color	Red; → Named colors #ff0000; → hexadecimal rgb(255,0,0); → RGB transparent; → transparent currentcolor (current text color)

Individual कोई भी element के specific section पर Border properties लगा सकते हैं

Border-top: 2px dashed blue; border-right: 3px solid green;, ……..
Border-top-width:5px; Border-top-style:solid; Border-top-color:black;
Border-radius: 10px;

```html
<style>
   .bihar{
    background-repeat :no-repeat;
     background-size:contain;
    width:150px;
  height:100px;
  border : 2px solid black;
  }
  p{
    border-top: 5px dotted red;
    border-left: 5px inset blue;
    border-bottom: 5px groove green;
    border-right: 5px ridge black;
  }
</style>
<body>

<p> Lorem ipsum dolor sit amet consectetur adipisicing elit. Iste quibusdam, nesciunt magnam sequi odit quis corrupti
praesentium enim velit facere mollitia aliquam minus eius! Beatae ipsum repellendus cumque! Eaque, mollitia?</p>
<a> Lorem ipsum dolor sit amet consectetur adipisicing elit. Ipsam quasi rerum neque explicabo magnam. Fugiat tempora
itaque, dicta error distinctio facilis commodi veritatis nesciunt officia. Cupiditate maxime magnam ratione deleniti.</a>
<div class="bihar">
</div>
```



5. Width property

- Width property is used to set the width of an element.
- Syntax → width: value;

Length/keyword Values		
Property	**Value**	**Work**
Width:	300px;	Fixed width in pixels
Width:	50%;	Percentage of parent container
Width:	Auto;	Browser calculates default width
Width:	100vw;	100% of viewport width
Width:	50rem;	Relative to root font-size
Width:	Fit-content;	Shrinks to fit content
Width:	Min-content;	Smallest possible width

Width:	Max-content;	Width based on content
Width:	Inherit;	Inherits from parent
Width:	initial	Sets to default (auto)

Absolute lengths/units		
Property	**Value**	**Work**
Width:	500px;	Fixed pixel width
Width:	10in;	Inches
Width:	20cm;	Centimeters
Width:	200mm;	Millimeters
Width:	12pt;	Points (1pt = 1/72 inch)
Width:	2pc;	Picas (1pc = 12pt)

Relative lengths		
Property	**Value**	**Work**
Width:	50%;	50% of parent's width
Width:	50vw;	50% of viewport width
Width:	50vh;	50% of viewport height
Width:	25em;	Relative to elements font-size
Width:	25rem;	Relative to root font-size
Width:	50vmin;	50% of smaller viewport dimension
Width:	50vmax;	50% of larger viewport dimension
Width:	100cqw	100% of container width

Content- based values		
Property	**Value**	**Work**
Width:	50%;	50% of parent's width
Width:	50vw;	50% of viewport width

Min/max & responsive image properties and values		
Property	**Value**	**Work**
Max-Width: Min-width: Width:	1200px; 300px; Clamp(200px,50%,800px);	Till 1200px in big screen Till 300px in small screen Flexible image size

Invalid ❌

Width:500; Width: 50%px; Width: auto 100px; Width: red; Width: calc(100%-);	Unit missing Double unit Multiple values Invalid value Incomplete calc

6. Height property

- Height property is used to set the height of an element.
- Syntax – height: value;

Example –

Keyword values

Property	Value	Work
Height:	Auto;	Default- browser calculates
Height:	Inherit;	Takes value form parent element
Height:	Initial;	Resets to default value (auto)
Height:	Unset;	Inherit if proper is inheritable
Height:	Revert;	Reverts to browser default styling
Height:	Revert-layer;	Reverts to previous cascade layer value

Length values

Absolute lengths		
Property	**Value**	**work**
Height:	100px;	Pixels
Height:	5cm;	Centimeters
Height:	5mm;	Millimeters
Height:	2in;	Inches
Height:	12pt;	Points
Height:	1pc;	Picas
Height:	10Q;	Quarter-millimeters
Relative lengths – fonts relative		
Height:	10em;	Relative to elements font size
Height:	10rem;	Relative to root elements
Height:	5ex;	Relative to height of “x” character
Height:	5ch;	Relative to width of”0” character
Height:	5Lh;	Relative to element’s line height
Relative lengths – Viewport		
Height:	50%;	Relative to parent element
Height:	100vh;	100% of viewport height
Height:	50vh;	50% of viewport height
Height:	100 svh	Small viewport height
Height:	5vmin	5% of the smaller dimension

Intrinsic sizing keywords

Property	Value	Work
Height:	Min-content;	Minimum height needed for content
Height:	Max-content;	Height needed for all content without wrapping
Height:	Fit-content;	Shrink to fit content, respects max-height
Height:	Stretch;	Stretches to fill container

CSS function

Property	Value	Work
Chooses the smallest value		
Height:	Min(500px,100vh)	500px or 100vh, whichever is smaller
Chooses the largest value		
Height:	Max(300px,50%)	300px or 50%, whichever is larger
Clamp() – sets min, preferred and max values		
Height:	Clamp(300px,50vh,800px);	b/w 300px-800px, prefers 50vh
Calc()-mathematical calculations		
Height:	Calc(100%-40px);	Full height minus 40px

Min/max & responsive image properties and values		
Property	**Value**	**Work**
Max-height: Min-height: height	500px; 200px; Clamp(200px,50%,800px);	Element won't be taller than 500px Element won't be shorter than 200px Flexible image size

Font-size property
Font-size property is used to change the size of text.
Text-align property
Text-align property is used to align text (left, right,center)

1. Background Property
Background property is used to set the background color, background image and so many function perform on an element.

```html
<title>hira css</title>
<style>
  div p{
    background-color   : yellow;
  }
  div a{
    color: white; background-color   : green;
  }
</style>
<body>
  <div>
    <p> Hello world !</p>
    <div>
      <div>
        <a> submit </a>
      </div>
</body>
```

http://127.0.0.1:5500/index.html

Hello world !

submit

```html
<title>hira css</title>
<style>
  *{
    background-color  : yellow ;
  }

</style>
<body>
  <div>
    <p> Hello world !</p>
    <div>
      <div>
        <a> submit </a>
      </div>
</body>
```

Hello world !

submit

```html
<title>hira css</title>
<style>
  *{
    background-image  : url(bihar.jpg) ;
    background-repeat : no-repeat
  }

</style>
<body>
    <p> Hello Bihar !</p>

</body>
```

Hello Bihar !

a. Background Color

Pro- It sets the background color of an element

Value- red, blue, green, #ffffff, rgb()

Example -

```
div{
        background-color: yellow;
}
```

http://127.0.0.1:5500/index.html

Hello hira

```html
<!DOCTYPE html>
<html>
<head>
<title>hira css</title>
<style>
.class{
background-color: yellow;
}
    </style>
    <body>
<div class="class">
<p> Hello hira </p>
</div>
    </body>
```

b. Background Image

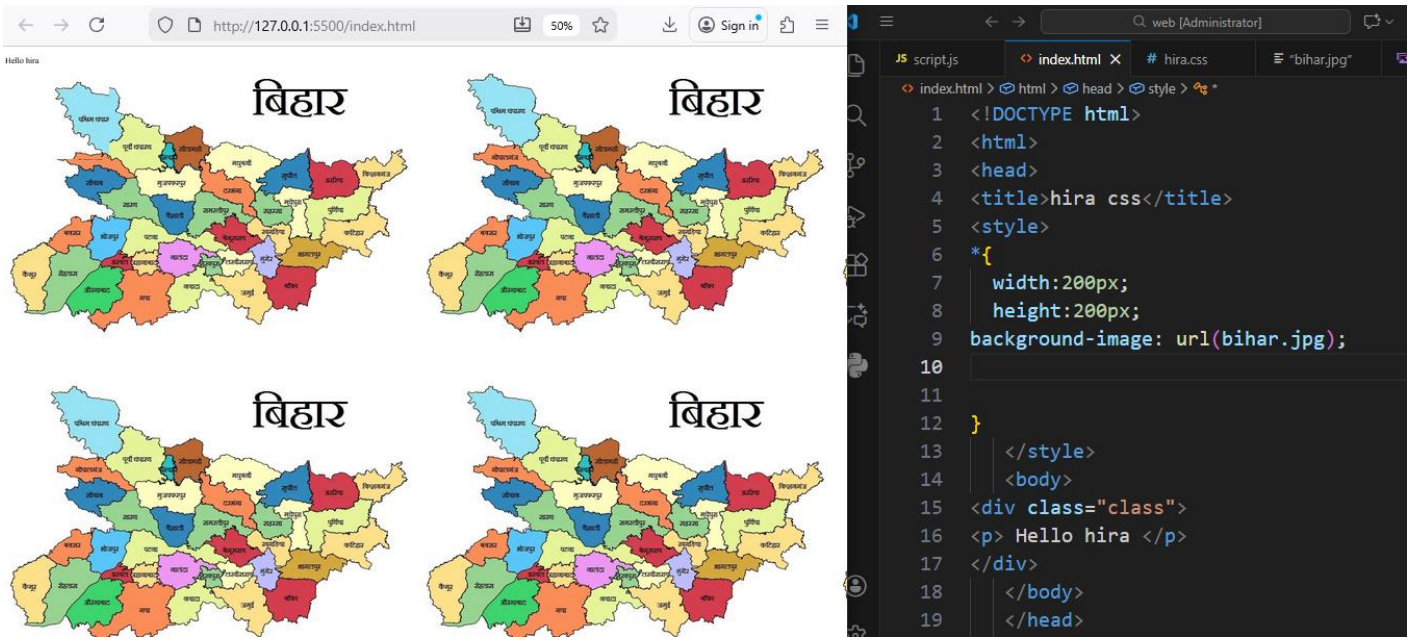
Pro- It sets an image as the background.

Value- url("image.jpg")

Example -

```
div{
        background-image:url("nature.jpg");
```

}



c. Background Repeat

Pro- It controls how the background image repeats.

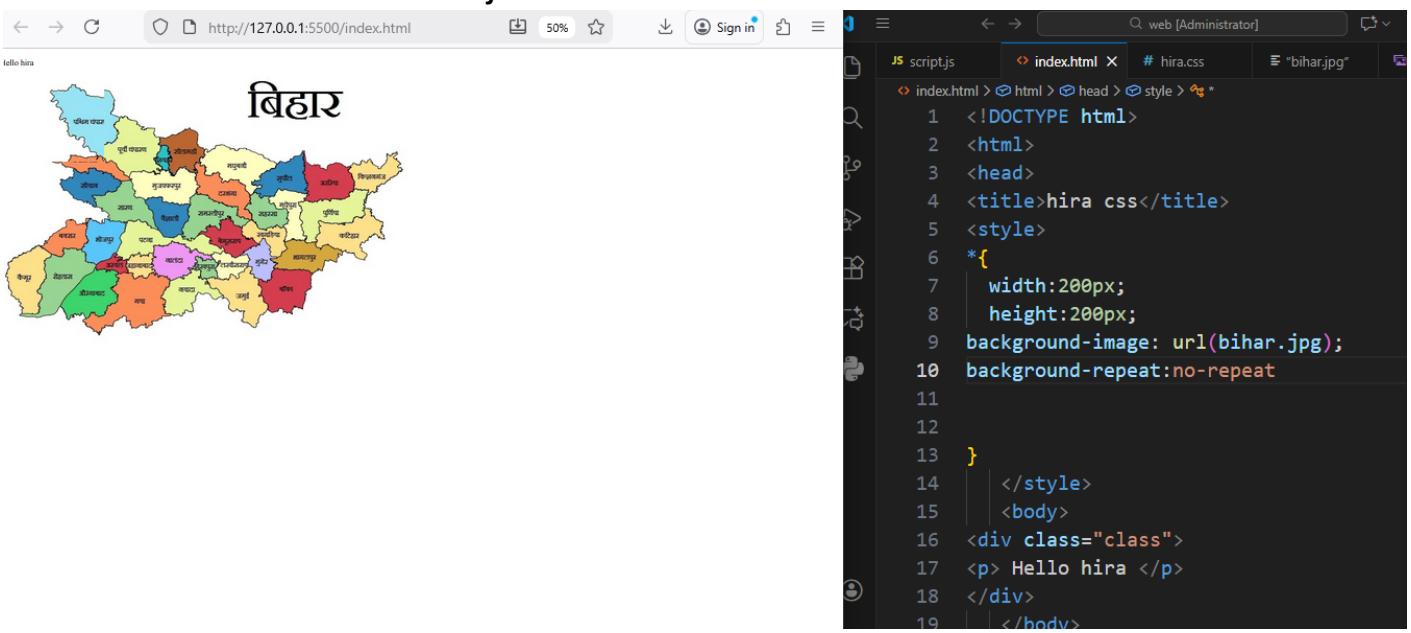
Value- repeat (by default), no-repeat, repeat-x, repeat-y

Example - div{

background-image:url("pattern.png");

background-repeat:no-repeat;

}



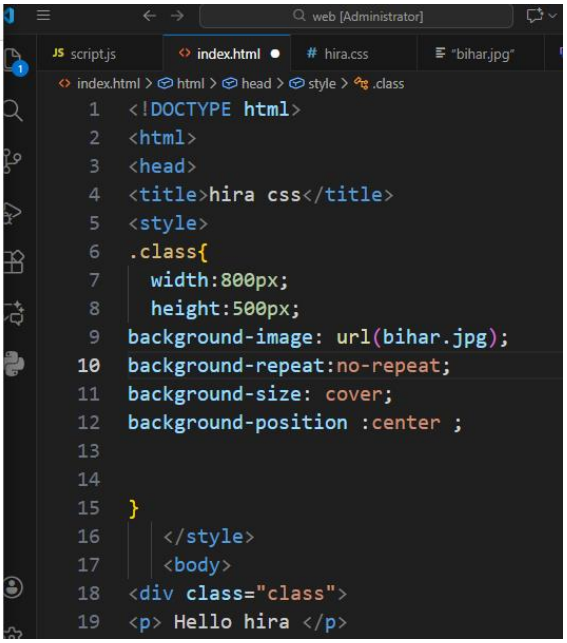
d. Background Size

Pro- It sets the size of the background image.

Value- auto, cover, contain (fit on box), 200px 150px

Example -

```
div{
        background-image:url("nature.jpg");
        background-size:cover;
        }
```



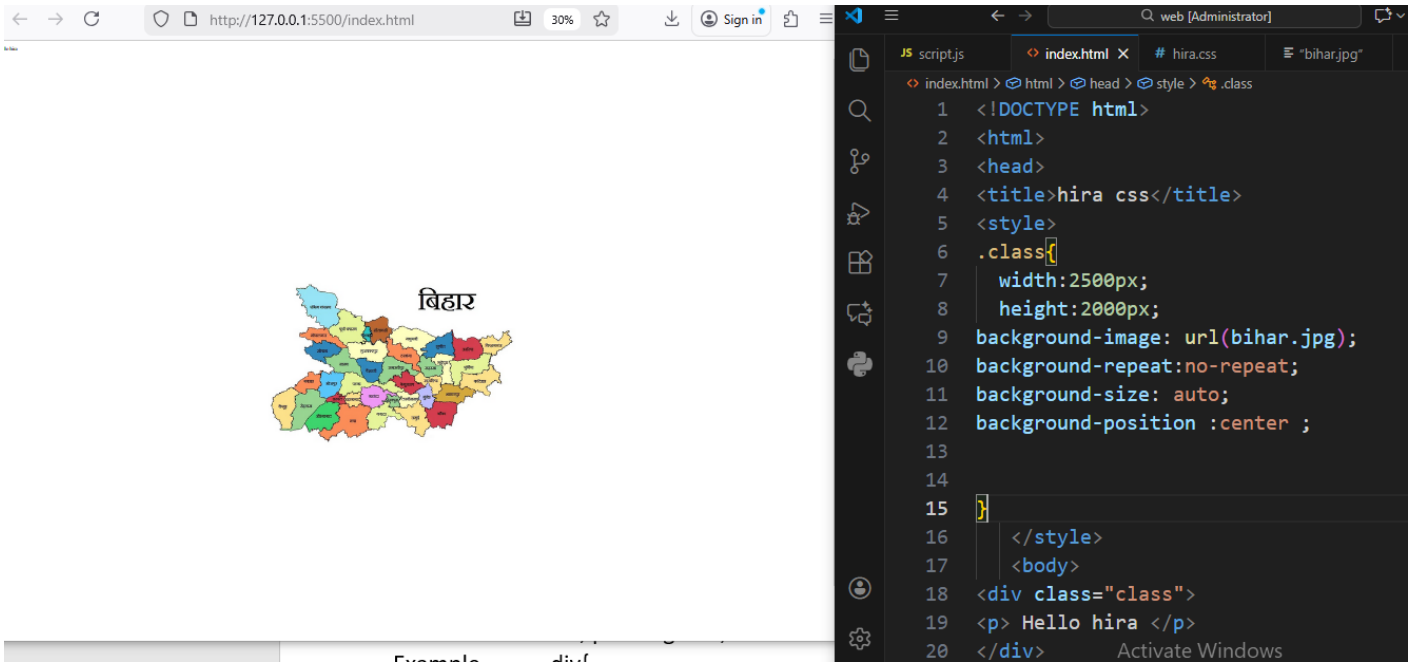
e. Background Position

Pro- It defines the starting position of the background image.

Value- left, right, center, top, bottom, 20px 30px

Example -

```
div{
        background-image:url("nature.jpg");
        background-position:center;
        }
```



f. Background Clip

Pro- it defines how far the background extends inside the element.

Value- border-box, padding-box, content-box

Example -

```
div{
    border:5px solid black;
    padding:20px;
    background-color:yellow;
    background-clip:content-box;
    }
```

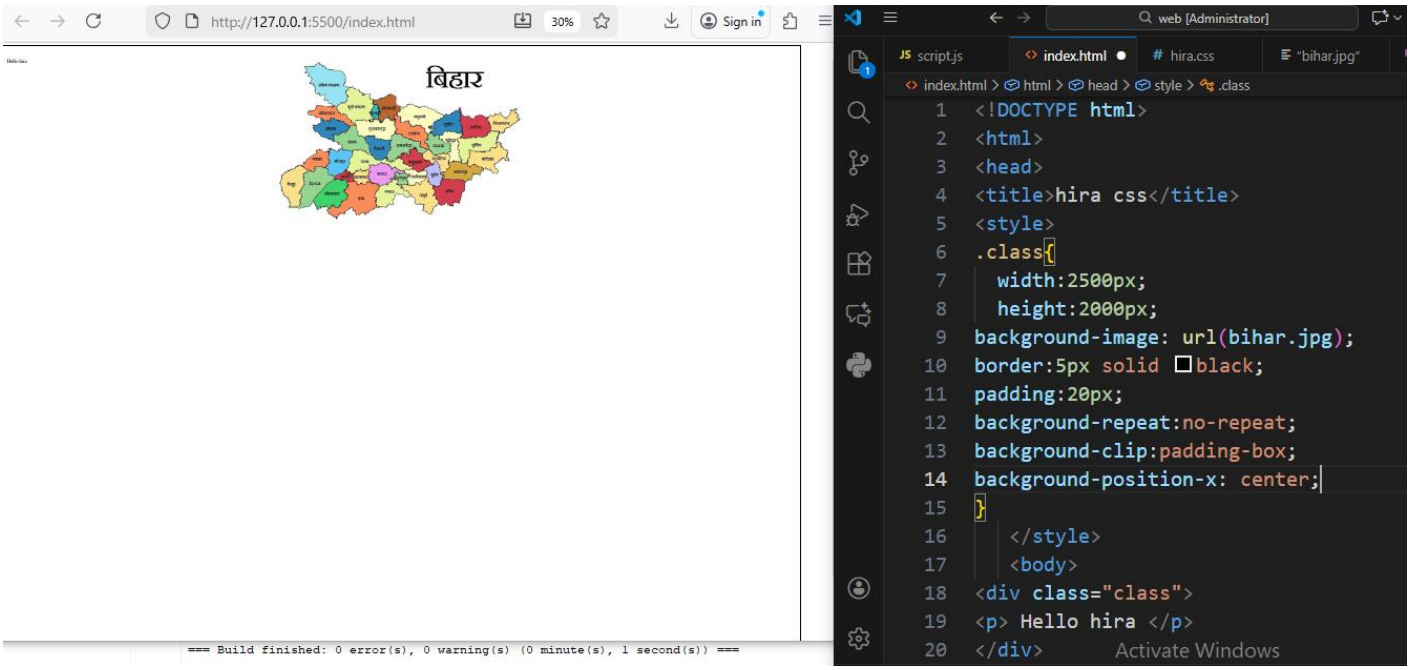
g. Background Position-X

Pro- It sets the **horizontal position** of the background image.

Value- left, center, right, 50px

Example –

```
div{
    background-image:url("nature.jpg");
    background-position-x:right;
    }
```



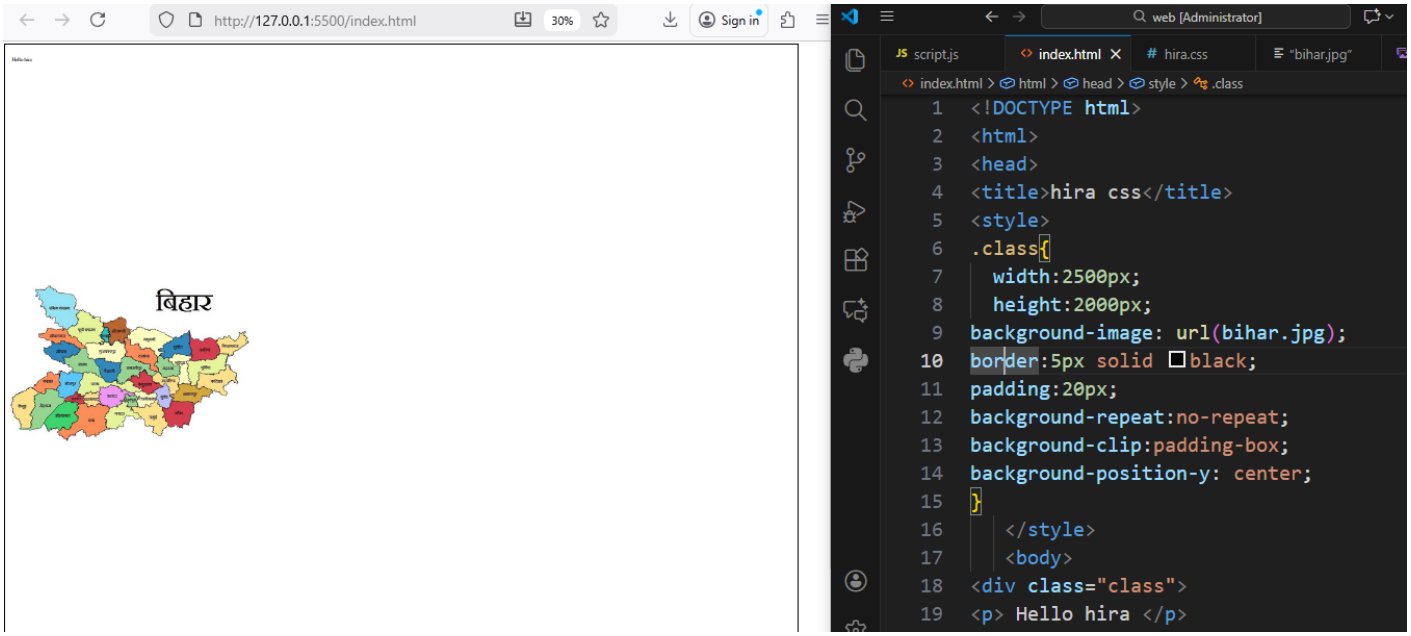
h. Background Position-Y

Pro- It sets the **vertical position** of the background image.

Value- top, center, bottom, 40px

Example -

```
div{
    background-image:url("nature.jpg");
    background-position-y:bottom;
}
```



i. Background Attachment

Pro- It controls whether the background image scrolls with the page or stays fixed.

Value- scroll, fixed, local

Example -
```
body{
    background-image:url("nature.jpg");
    background-attachment:fixed;
    }
```

j. Background Origin

Pro- It defines where the background image positioning area starts.

Value- border-box, padding-box, content-box

Example -
```
div{
    padding:30px;
    border:5px solid black;
    background-image:url("nature.jpg");
    background-origin:padding-box;
    }
```

k. Background Blend Mode

Pro- It blends the background image with the background color.

Value- multiply, overlay, darken, lighten

Example -
```
div{
    background-image:url("nature.jpg");
    background-color:red;
    background-blend-mode:multiply;
    }
```

l. Shorthand → all view write in one line of background. (e.g. → color, image, repeat, attachment, position)

Syntax → background: color image repeat attachment position;

2. Text & Font properties

<style> p{	Work
color: red;	Text color
font-size: 16px;	Text size

font-style: italic;	Text make italic.
font-weight: 500;	Text bold (range 100-900)
font-family: arial;	Text font type
text-align :center;	Text alignment
text-decoration: underline	Text underline
line-height: 1.5;	Gap b/w two text
letter-spacing: 2 px;	Gap b/w two letter
word-spacing: 10px	Gap b/w two words
text-transform: uppercase;	Text uppercase/lowercase
text-indent: 50px;	indent the first line of text.
Syntax - text-shadow: horizontal vertical blur color; Example - text-shadow: 2px 2px 5px gray;	add shadow to text.
} </style>	

Note :- apply multiple font properties in one line.

(आप एक से ज्यादा font property को एक line में use कर सकते हैं)

Syntax → font: font-style font-weight font-size font-family;

Example → font: italic bold 20px Arial;

a. Text & Font properties with value

Properties	Value
Color:	Red, green,yellow, white,sky blue, dark bule,…..
Font size:	px, em, rem, %
Font style:	normal, italic, oblique
Font weight:	normal, bold, lighter, 100 – 900
font-family:	Arial, Times New Roman, Verdana, Georgia
text-align:	left, right, center, justify
text-decoration:	none, underline, overline, line-through
text-decoration-style:	Dashed, double,solid,wavy

Text-decoration-color:	Red,blue,green,…….
line-height:	px
letter-spacing:	px
text-transform:	uppercase, lowercase, capitalize
text-indent:	Px
text-shadow:	px color

Example –

```
<!DOCTYPE html>
<html>
<head>
<style>
h1{
  text-align:center;
  text-decoration:underline;
  text-shadow:2px 2px 5px gray;
  font-size:100px;
  color: red;
}
p{
  font-family:Arial;
  font-size:20px;
  font-style:italic;
  font-weight:bold;
  letter-spacing:2px;
  word-spacing:8px;
  line-height:30px;
}
</style>
</head>
<body>
<h1>Hello World !</h1>
```

This paragraph shows text and font properties.
</body>
</html>

```
<head>
<style>
h1{
  text-align:center;
  text-decoration:underline;
  text-shadow:2px 2px 5px gray;
  font-size:100px;
  color: red;
}
p{
  font-family:Arial;
  font-size:20px;
  font-style:italic;
  font-weight:bold;
  letter-spacing:2px;
  word-spacing:8px;
  line-height:30px;
}
</style>
</head>
<body>
<h1>Hello World !</h1>
<p>This paragraph shows text and font properties.</p>
</body>
```

http://127.0.0.1:5500/index.html

Hello World !

This paragraph shows text and font properties.

b. Dimension Properties

Properties: Value	Work

Width: 200px Width: 10rem Width: 50%	Fixed pixel width Relative to root font-size Percentage of parent container

c. Background Properties

<syntax> div{	Work
Background-color : lightblue	Background color
Background-image: url(“pic.jpg”)	Background image
Background-size:cover;	Image size
Background-position: center;	Image position
Background-repeat: no-repeat;	Repeat or not

Value

Property	Value & work
Background-color:	Red; → Named colors #ff0000; → hexadecimal rgb(255,0,0); → RGB transparent; → transparent currentcolor (current text color)
Background-image:	url(“image.jpg”) → Single image url(“pic1.jpg),url(pic2.jpg),… → multiple image None; → no image Linear-gradient (red,blue); →gradient

HIRA KUMAR